School of Biomedical Engineering, Science & Health Systems

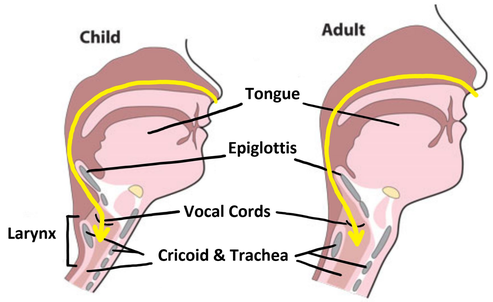
Pediatric Nasotracheal Intubation Assist Device

Casey Fuoco[1], Sophia Macchia[1], Hannah Mikkelsen[1], Dominic Razo-Castañeda[1], Sarthak Sharma[1], Marek Swoboda, PhD[1], Aysha Hasan, MD[2]

[1] School of Biomedical Engineering, Science and Health Systems [2] St. Christopher's Hospital for Children

Background

- Delays in **airway management** increase mortality rate from **1.8%** to **11.8%** in hospital and emergency settings[1]
- Oral **endotracheal intubation** is most common solution method
- **57%** of complications in **pediatric** oral intubation cases involve **pre-existing airway or craniofacial abnormalities** requiring a **different method**[2]



Alternative Path: Major differences in pediatric and adult nasopharyngeal pathway

Existing Solutions & Limitations

Nasal Trumpet (Nasopharyngeal Airway)

Pros	Cons
Opens obstructed airway	Prevents insertion of endotracheal tube
Guides fiberscope (off label use)	Increases time needed to establish airway (30-40 minutes)
Non-Invasive	Requires manual cut while in use- risk of human error

Constraints

Project Constraints:

1. Time - 9 Months
2. Budget - 800 USD
3. Resources - St. Christopher's Hospital
4. Standards - ISO 10993, ASTM D624, ASTM D3767-03

Design Constraints:

1. Patient Anatomy
2. Disruption of Normal Procedure
3. Patient Safety

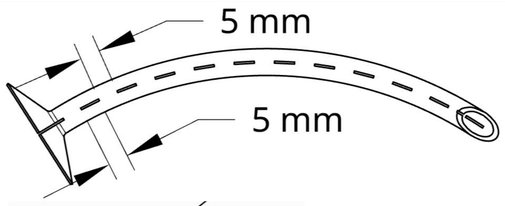
Requirements

R1. Outer Diameter= **6.7 ± 0.2 mm**
R2. Inner Diameter = **5.0 ± 0.2 mm**
R3. Length = **105 ± 0.2 mm**
R4. Tear Strength = **14.3N**
R5. Flexibility = Bending Stiffness must **not be significantly different** from existing trumpets ($4.5*10^{-4}$ Nm^2) at 5% significance level

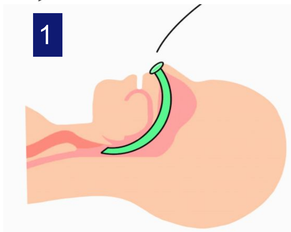
Objective

Create an **easily removable** assistive device for **pediatric nasal** fiberoptic intubation especially in challenging, difficult airways.

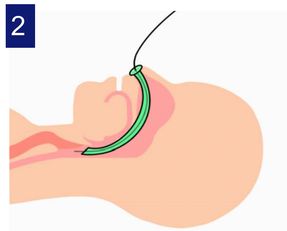
Solution

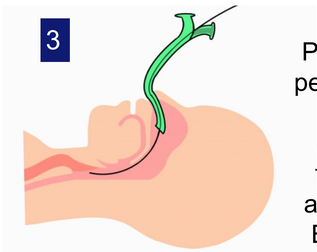


Engineering Drawing: with same dimensions as existing (20 French) trumpet and added perforations

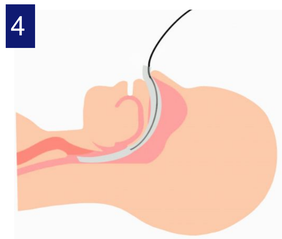


The nasal trumpet placed and is able to properly guide the fiberscope to the correct location.



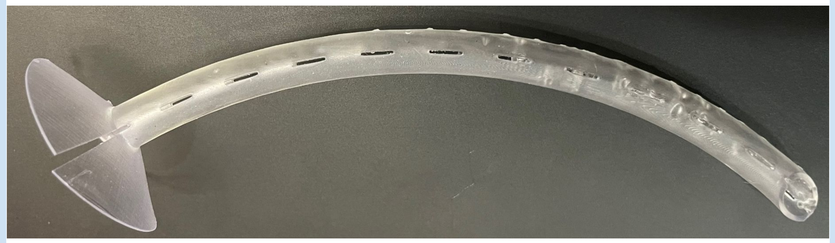


Pre-manufactured perforations ensure the trumpet is easily removed from the patient and allows for the ET to be placed.



Working Models & Prototype

Prototype: 3D printed in Flexible Material using Stereolithography (SLA) printer

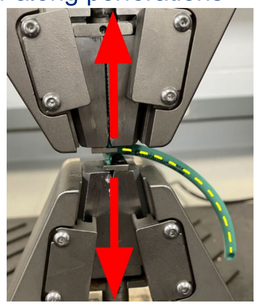


Future Revision: Will be injection molded using medical grade neoprene to ensure biocompatibility.

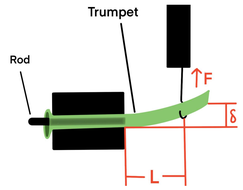
Verification Testing

Measurements (R1 - R3): Measure outer diameter, inner diameter and length with caliper

Tensile tearing test (R4):
To find the max force required to tear along perforations



Bending stiffness test (R5):
Cantilever bending test for non-perforated and perforated trumpet
Compared with 2 tailed t-test



$$Bending\ Stiffness = \frac{FL^3}{3\delta}$$

R1: OD	R2: ID	R3: Length	R4: Tear Strength	R5: Flexibility
6.7± 0.2 mm	5 ± 0.2 mm	105 ± 0.2 mm	<14.3 N	($4.5*10^{-4}$ Nm^2)
✓	✓	✓	=13.5N	($3.4*10^{-4}$ Nm^2) p=0.09

Simulated-Use Testing: Device was successful in a simulated procedure using mannequins at St. Christopher's Children's Hospital.

Societal Impact

Will benefit physicians, anesthesiologists, and patients by improving nasal intubation via fiberoptic guidance in emergencies.

Conclusion

This device will assist in difficult pediatric nasotracheal intubation by reducing the risk of critical delays in airway management in cases where the mouth is inaccessible.

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[1] B. Morshedi, "Management of the trauma patient's airway – pearls and pitfalls," (2015)
[2] Bai W, et al.; "Evaluation of emergency pediatric tracheal intubation by pediatric anesthesiologists on inpatient units and the Emergency Department," (2016)
[3] ASTM International. "ASTM *D3767-03 Standard Practice for Rubber—Measurement of Dimensions*," 2010.
[4] ASTM International. "ASTM D624-00(2020) *Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers*